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THE COPPERBELT UNIVERSITY

SCHOOL OF MATHEMATICS AND NATURAL SCIENCES

COMPUTER SCIENCE DEPARTMENT

CS235 Database technology TEST 2

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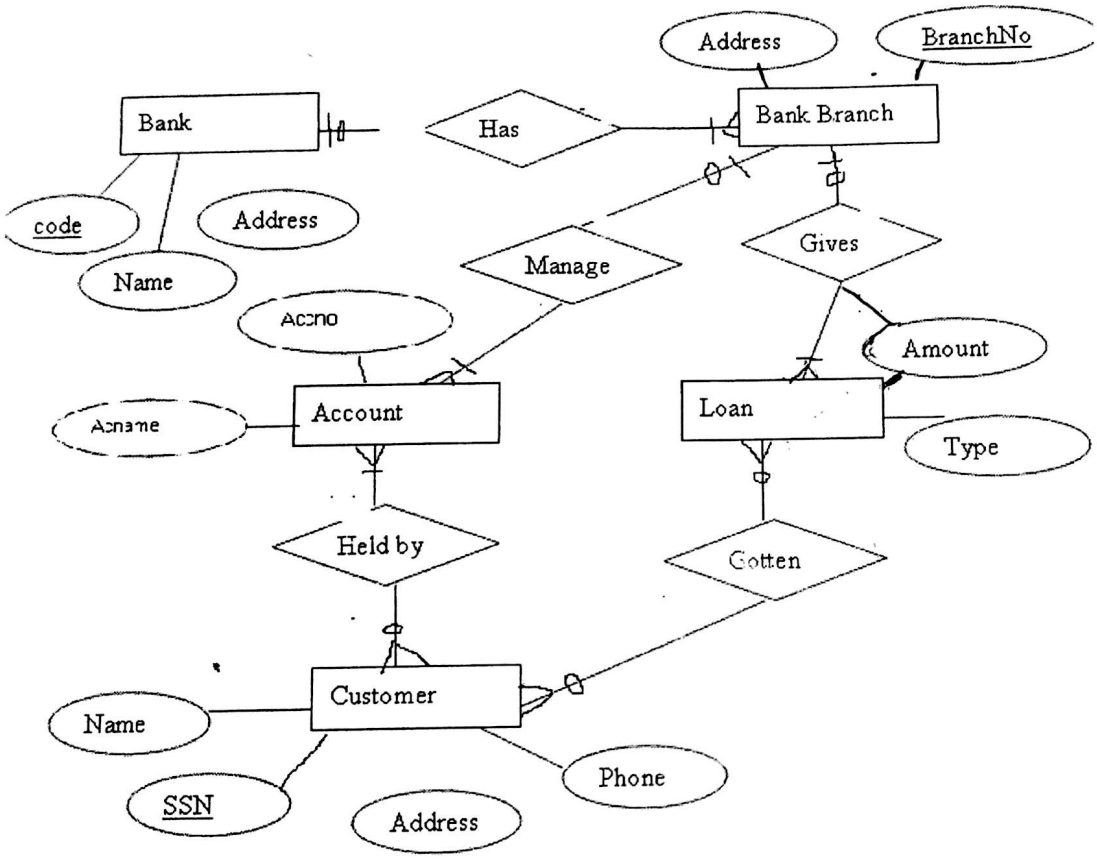
QUESTION ONE

Map the ER Diagram to relational schemas showing all the necessary steps. **[8 marks]**



QUESTION TWO

The table below is susceptible to update anomalies. Provide from the table, examples of insertion, deletion, and modification anomalies. **[6 marks]**

EmpNo	ContractNo	HoursPerWeek	eName	ProjectNo	ProjectLocation
3121	C201	16	James Banda	P25	Lusaka
3245	C201	24	Mary Musonda	P25	Lusaka
2345	C202	28	Martha Mwale	P20	Kitwe
3121	C202	16	James Banda	P20	Kitwe

QUESTION THREE

When is a table said to be in second normal form (2NF) and Third normal form (3NF) respectively and given a relation R = {A, B, C, D, E, F, G, H, I,J} and the set of functional Dependencies. F ={{A, B} → {C}, {A} → {D, E}, {B} → {F}, {F} →{G, H},{D}→ {I, J }}, Determine the key for R and hence or otherwise decompose R into 2NF and then 3NF relations.

[8 marks]

QUESTION FOUR

Consider the following relational schema:

employee (ssn, name, gender, address, salary, supervisorSSN, dnumber)

department (dnumber, dname, managerSSN)

deptLocation (dnumber, dlocation)

project (pnumber, pname, plocation)

worksOn (emplSSN, pnumber, hours)

Answer these queries using both relational algebra and SQL statements: **[8 marks]**

i). Get the ssn of employees whose salaries are greater than the average salary.

ii). Find department names for departments with more than 30 employees

iii). Find the number of employees working on projects located in Kitwe.

iv). Find the number of employees and the total salary for employees in each department

v). Find the supervisor's social security numbers for supervisors supervising more than 10 employees

QUESTION FIVE

a) State and describe the four (4) types of fragmentation. **[4 marks]**

b) State and describe the Four (4) alternative strategies regarding placement of data.

[4 marks]

.....THE END.....